

# RIGHT-SIDED HEART DISORDERS

Michael Ragosta

After Werner Forssmann boldly inserted a urological catheter into his own right atrium, the right heart became accessible to clinical investigation, allowing the study of right heart physiology in both normal and diseased states.<sup>1</sup> The right heart is affected by many cardiac pathologies. Recall that the primary cause of right heart failure is left heart failure; therefore the myriad cardiac disorders associated with left heart failure syndromes often impact right heart hemodynamics. In addition, numerous congenital heart conditions, as well as disorders of the pericardium, affect right heart hemodynamics. The influences these conditions have on right heart hemodynamics are discussed in their respective chapters. This chapter will focus on disorders unique to the right heart, including tricuspid and pulmonic valvular diseases and the hemodynamics of right ventricular failure with a focus on right ventricular infarction.

## TRICUSPID VALVE STENOSIS

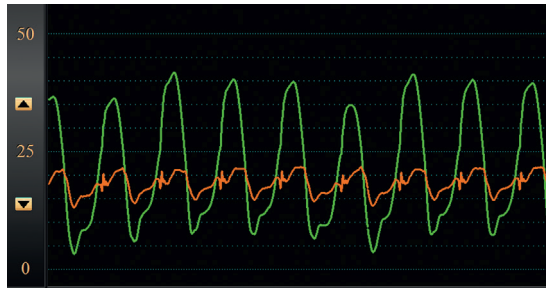
This rare valvular lesion is most often due to rheumatic heart disease and is almost always associated with mitral stenosis; isolated rheumatic tricuspid stenosis is very rare.<sup>2</sup> Only occasionally is tricuspid stenosis caused by other conditions, including carcinoid syndrome, endomyocardial fibrosis, congenital tricuspid valve stenosis, endocarditis, pacemaker lead–related leaflet fibrosis, or atrial myxoma. In the current era the most commonly observed cause of tricuspid stenosis is from structural deterioration and stenosis of a surgically inserted prosthetic tricuspid valve. With the development of novel therapies for the percutaneous treatment of tricuspid regurgitation such as edge to edge repair, stenosis may occur as a consequence of these therapies.<sup>3</sup>

Tricuspid stenosis impairs right atrial emptying and elevates right atrial pressure. Diminished filling of the right ventricle reduces cardiac output. In cases of rheumatic heart disease, the combination of tricuspid and mitral stenosis reduces the cardiac output to levels lower than expected on the basis of either valvular lesion alone. Clinical consequences of severe tricuspid stenosis include fatigue caused by low cardiac output, elevated jugular veins, peripheral edema, hepatic congestion, and ascites due to elevated right atrial pressure. If unsuspected, the diagnosis may prove challenging because these symptoms occur in other conditions such as pericardial disease, cirrhosis of the liver, and pulmonary hypertension; the latter may, in fact, be present due to associated mitral stenosis.

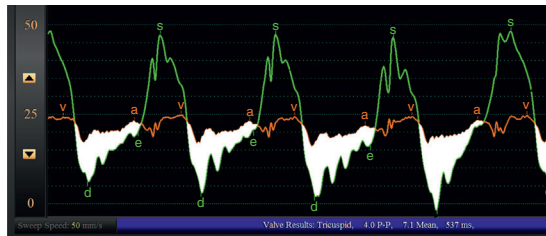
The hemodynamic abnormalities observed in tricuspid stenosis have been well described.<sup>4–7</sup> Right atrial pressure is elevated. The *a* wave reaches giant proportions in patients with normal sinus rhythm and may exceed 20 mm Hg. However, an enlarged *a* wave is not specific for tricuspid stenosis because it may be seen in the presence of pulmonary hypertension and right ventricular hypertrophy; although, in the absence of pulmonary hypertension, a prominent *a* wave supports a diagnosis of tricuspid stenosis.

Similar to mitral stenosis, the presence of a pressure gradient observed while simultaneously measuring pressure in the right atrium and right ventricle during diastole characterizes tricuspid valve stenosis (Fig. 8.1). Because of the lower right-sided pressures, the relatively lower cardiac output, and the greater size of the tricuspid orifice when compared with the mitral valve, the observed gradients are correspondingly relatively small, ranging from 2 to 12 mm Hg, with 90% of gradients less than 7 mm Hg.<sup>7</sup> A mean diastolic gradient >2 mm Hg is diagnostic of tricuspid stenosis. Small gradients (2–3 mm Hg) that exist only in early diastole may be observed in patients with predominantly tricuspid regurgitation without significant stenosis.<sup>4,7</sup> In patients with tricuspid stenosis and normal sinus rhythm, a small pressure gradient early in diastole increases at the end diastole because of the rise in atrial pressure from atrial contraction. For patients with atrial fibrillation, right atrial pressure remains uniformly elevated throughout the cardiac cycle, and the pressure gradient is greatest in early diastole when the right ventricular diastolic pressure is lowest. The transtricuspid valve pressure gradient increases with inspiration, predominantly caused by a fall in the ventricular diastolic pressure with inspiration. The gradient increases with exercise because of an increase in the right atrial pressure. An increase in volume will also increase the gradient.

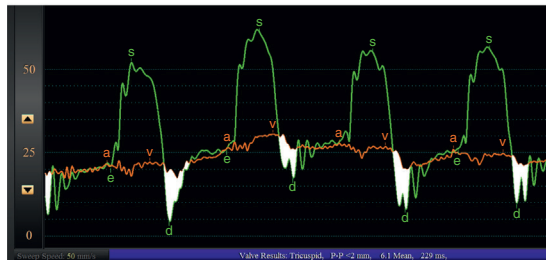
Calculation of the tricuspid valve orifice area has been estimated using the Gorlin formula (see Chapter 5). Similar to mitral stenosis, the mean pressure gradient across the valve, the diastolic filling period, the heart rate, and the cardiac output are the important measured variables entered into the formula; however, unlike the mitral valve, the coefficient has not been determined and has been arbitrarily set at 1.0 (similar to the aortic valve area). The formula has not been well validated in tricuspid stenosis, although small series have correlated the calculated valve area with the area determined at surgery.<sup>7</sup> Similar to mitral stenosis with associated mitral regurgitation, if there is associated tricuspid regurgitation, the Gorlin formula will underestimate the valve area because the true transvalvular flow is not known.



**Fig. 8.1** These hemodynamic waveforms were obtained from a 46-year-old male with a history of congenital ventricular septal defect repair at age 8 and subsequent tricuspid valve replacement for severe tricuspid valve regurgitation at age 17, who then developed severe stenosis of the bioprosthetic tricuspid valve. Simultaneous right atrial and right ventricular pressure waveforms are shown. The right atrial pressure is markedly elevated, and there is a large diastolic pressure gradient.



A



B

**Fig. 8.2** The tracings shown here were obtained in a 50-year-old male with a history of traumatic tricuspid valve injury who had a bioprosthetic tricuspid valve replacement 19 years earlier. (A) Simultaneous right atrial and right ventricular pressure waveforms confirmed severe tricuspid stenosis. He then underwent a percutaneous valve-in-valve procedure using a 26-mm SAPIEN 3 valve. (B) Postvalve implantation, there is no significant end-diastolic gradient present. The white shaded area represents the pressure gradient during diastole. *a*, a wave; *d*, diastole; *e*, end diastole; *v*, v wave.

The value of this determination is not clear, and today most assessments of the severity of tricuspid stenosis are made based on the extent of the transvalvular gradient and its effect on right atrial pressure.

Until recently, stenosis of a prosthetic tricuspid valve could only be treated surgically with another valve replacement. In the current era, less invasive alternatives are available, and transcatheter valve systems, initially designed for the aortic valve, have been employed to treat bioprosthetic tricuspid stenosis using a “valve-in-valve” approach. Case reports and small series demonstrate the feasibility of this approach with an improvement in hemodynamics.<sup>8–11</sup> Fig. 8.2 is an example of the hemodynamics obtained in a patient with bioprosthetic tricuspid stenosis treated with a SAPIEN valve (Edwards Lifesciences, Irving, California). In this case the gradient was reduced from a mean of 7.1 mm Hg (Fig. 8.2A) to 0 mm Hg at end-diastole postvalve deployment (Fig. 8.2B). While no postimplant gradient was seen in this case, residual gradients after valve-in-valve procedures for bioprosthetic tricuspid stenosis are common and may limit this approach except in patients at prohibitive risk for reoperation. Transcatheter “edge-to-edge” repair using a percutaneously deployed clip to reduce tricuspid regurgitation may result in small tricuspid valve gradients, with one study finding that nearly 5% of patients had more than a 5 mm Hg gradient after edge-to-edge repair.<sup>3</sup> The clinical significance of this finding is unclear at this time but certainly warrants caution.

## TRICUSPID VALVE REGURGITATION

Tricuspid regurgitation represents the most commonly encountered right-sided valvular heart lesion. Mild-to-moderate degrees of tricuspid regurgitation are very commonly detected on two-dimensional echocardiography and are of little to no significance. However, severe tricuspid regurgitation is a much more serious condition, affecting over 1.6 million Americans and causing progressive right heart failure and increased morbidity and mortality.<sup>12–14</sup> Among the numerous possible etiologies (Box 8.1), functional tricuspid regurgitation from right ventricular pressure or volume overload accounts for most cases; primary regurgitation caused by organic tricuspid valve pathology is much less prevalent.<sup>13,14</sup> In patients with rheumatic heart disease, tricuspid regurgitation is common, with a prevalence of nearly 40% in patients with mitral stenosis. Tricuspid regurgitation is due to several potential mechanisms in these patients, including rheumatic involvement of the tricuspid valve (i.e., primary tricuspid regurgitation) or functional regurgitation as a consequence of pressure or volume overload of the right ventricle (often caused by associated pulmonary hypertension).<sup>13,14</sup> Elucidation of the mechanism of tricuspid regurgitation seen in association with rheumatic mitral stenosis is important for proper treatment. Observation of normal pulmonary pressures suggests primary valve disease. In patients with pulmonary hypertension, echocardiography can help distinguish functional regurgitation from organic tricuspid valve disease.<sup>13,14</sup>

Tricuspid regurgitation causes volume overload in the right ventricle and atrium. Over time, the right ventricle dilates further, worsening the degree of regurgitation. In the presence of pulmonary hypertension, severe tricuspid regurgitation causes both volume and pressure overload and is less well tolerated, leading to an earlier onset of symptoms. Signs of severe tricuspid regurgitation reflect right heart failure and include lower extremity edema, ascites, distended neck veins, and cachexia. Symptoms include dyspnea, anorexia, abdominal distension, and profound fatigue from decreased cardiac output.

The observed hemodynamic abnormalities of severe tricuspid regurgitation are preload and afterload dependent and include elevation of right atrial pressure, decreased cardiac output, and abnormalities of the right atrial pressure waveform.<sup>15</sup> Because the jugular veins mirror the abnormalities present in the right atrium, it is no surprise that the characteristic atrial waveform abnormalities attributed to tricuspid regurgitation were first observed on analysis of jugular venous pressure waveforms (Fig. 8.3).<sup>15</sup> Normally, an *x* descent exists on the right atrial waveform, reflecting the descent of the base of the heart during systole. Classically, in tricuspid regurgitation, the *x* descent is attenuated (Fig. 8.4). The *x* descent ultimately disappears and is replaced by a systolic wave with a *peak-dome* contour often termed the *c-v* wave.<sup>1,16–18</sup> The *v* wave is classically prominent, and the *y* descent is very rapid (Fig. 8.5). Ventricularization of the right atrial pressure waveform may occur (Fig. 8.6). In some cases, the right atrial pressure wave is nearly indistinguishable from the right ventricular pressure contour (Fig. 8.7). The *v* wave may increase further during exercise.

Unfortunately, these hemodynamic findings are not always helpful in diagnosing tricuspid regurgitation. The findings are dependent on the preload and afterload status; thus they may be absent in the event of excessive diuresis or more prominent when volume is overloaded. Atrial fibrillation without tricuspid regurgitation may distort the atrial waveform in a fashion similar to that observed with severe regurgitation, with the absence of the *x* descent (because no atrial contraction is present) causing the *c-v* wave to appear prominent.<sup>18,20</sup> It is important to note that observation of a normal right atrial pressure, and normal *x* descent along with the absence of prominent *v* waves, does not necessarily exclude significant tricuspid regurgitation.<sup>20–22</sup> Ventricularization of right atrial pressure is very specific for severe

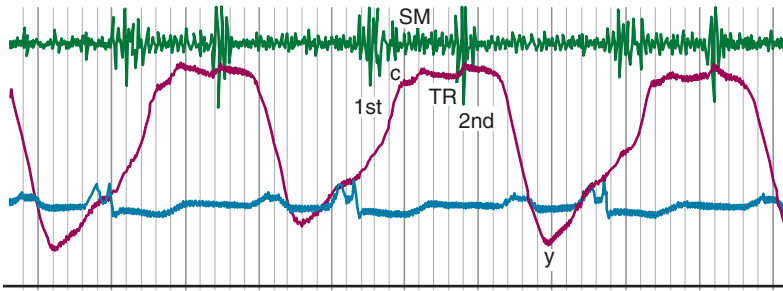
### Box 8.1 Causes of Tricuspid Regurgitation

#### Structurally Normal Tricuspid Valve (Functional Tricuspid Regurgitation)

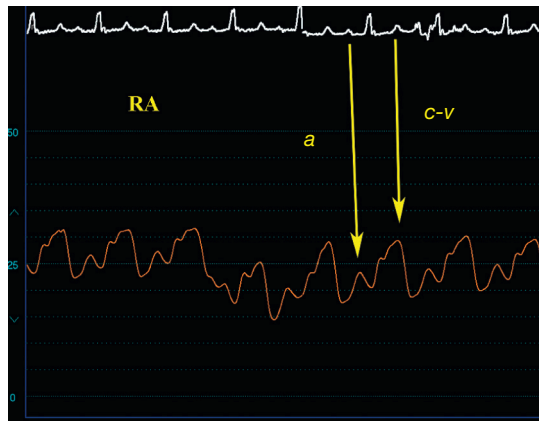
- Chronic atrial fibrillation
- Annular dilatation from volume or pressure overload
  1. Atrial septal defect
  2. Right ventricular infarction
  3. Congestive heart failure
  4. Pulmonary hypertension
  5. Postheart transplantation

#### Structurally Abnormal Tricuspid Valve

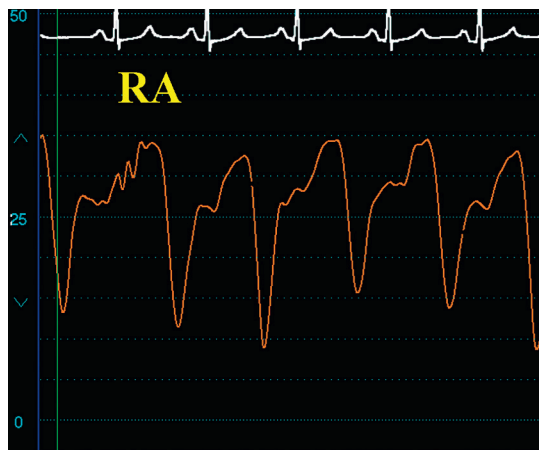
1. Rheumatic heart disease
2. Ebstein anomaly
3. Congenital heart disease
4. Carcinoid
5. Radiation-induced valvular regurgitation
6. Endocarditis
7. Trauma
8. Right ventricular biopsy induced
9. Pacemaker-lead induced
10. Myxomatous degeneration



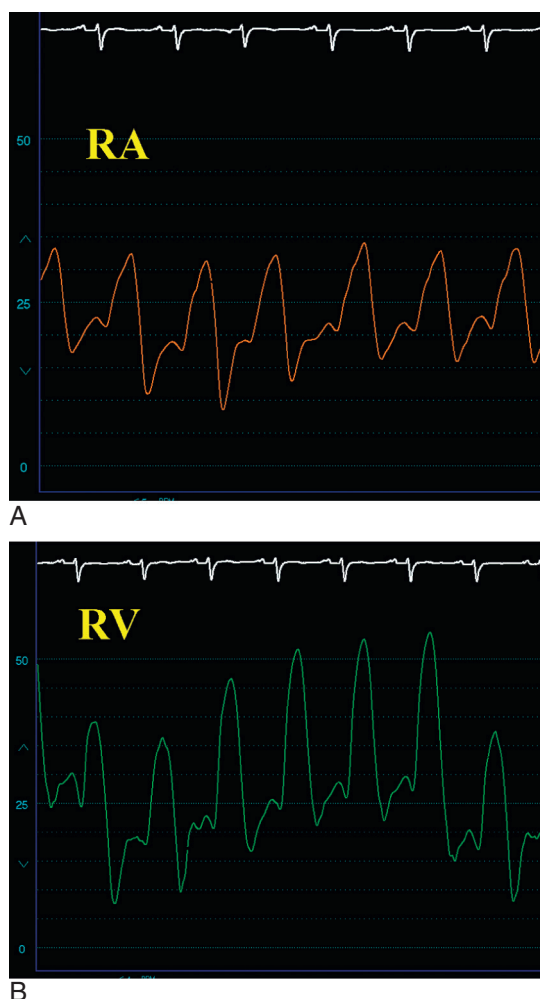
**Fig. 8.3** Venous pressure waveform in severe tricuspid regurgitation, demonstrating a large c-v wave.<sup>19</sup> (From Messer AL, Hurst JW, Rappaport MB, Sprague HB. A study of the venous pulse in tricuspid valve disease. *Circulation*. 1950;1:388–393) c, c wave; y, y descent.



**Fig. 8.4** Right atrial waveform from a patient with secondary tricuspid regurgitation from associated severe left-sided heart failure and right-sided heart failure. Attenuation of the x descent is present, seen after the a wave (left arrow), leading to a prominent c-v wave (right arrow). a, a wave; RA, right atrial.



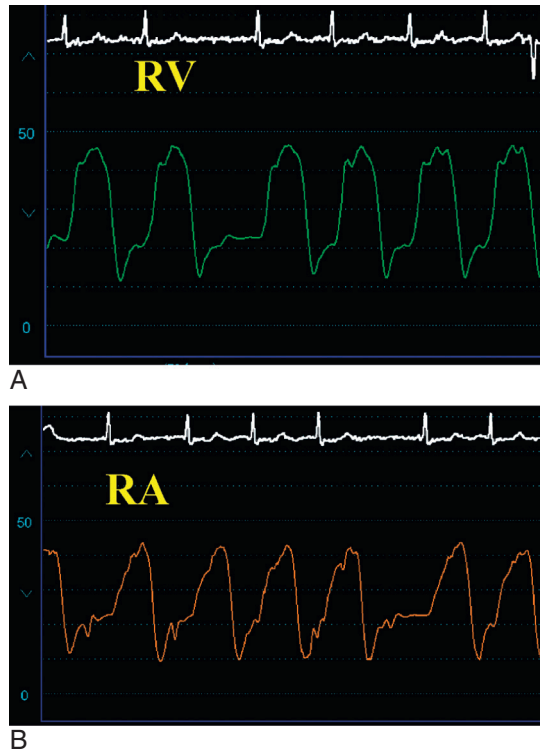
**Fig. 8.5** Right atrial waveform in severe tricuspid regurgitation, demonstrating the absence of the x descent and a large c-v wave with a prominent y descent. RA, Right atrial.



**Fig. 8.6** These tracings were obtained from a patient with severe tricuspid regurgitation due to profound biventricular heart failure. (A) The right atrial waveform shows tricuspid regurgitation. Compare this with (B), the right ventricular waveform from the same patient. *RA*, Right atrial; *RV*, right ventricular.

tricuspid regurgitation but is seen in only 40% of patients.<sup>22</sup> Similar to mitral regurgitation, the size of the v wave in tricuspid regurgitation depends on the volume status and compliance of the right atrium and does not necessarily correlate with the presence or severity of tricuspid regurgitation.<sup>23</sup> A subtle hemodynamic finding is perhaps more sensitive for tricuspid regurgitation. Instead of the normal fall in right atrial pressure with inspiration, one study found that all patients with tricuspid regurgitation demonstrated either a rise or no change in right atrial pressure during deep inspiration.<sup>22</sup> This finding was very sensitive for tricuspid regurgitation but was also apparent in several patients with severe (>90 mm Hg) pulmonary hypertension; thus in the absence of severe pulmonary hypertension, this sign may help diagnose tricuspid regurgitation. In contrast to mitral regurgitation, angiographic assessment of tricuspid regurgitation is problematic and rarely used, because the presence of a catheter across the tricuspid valve to perform right ventriculography may interfere with tricuspid valve function and cause regurgitation; however, this method may be useful for proving the absence of tricuspid regurgitation.

Some of the clinical and hemodynamic aspects of severe tricuspid regurgitation may be confused with constrictive pericarditis.<sup>24</sup> The predominant symptoms of both conditions (edema, ascites, prominent neck veins, and fatigue) are very similar. In addition, the right atrial pressure waveform may appear similar with a prominent y descent, particularly



**Fig. 8.7** Severe tricuspid regurgitation may result in complete ventricularization of the right atrial waveform. (A) The right ventricular pressure wave. (B) Note the nearly indistinguishable appearance of the right atrial waveform. RA, Right atrial; RV, right ventricular.

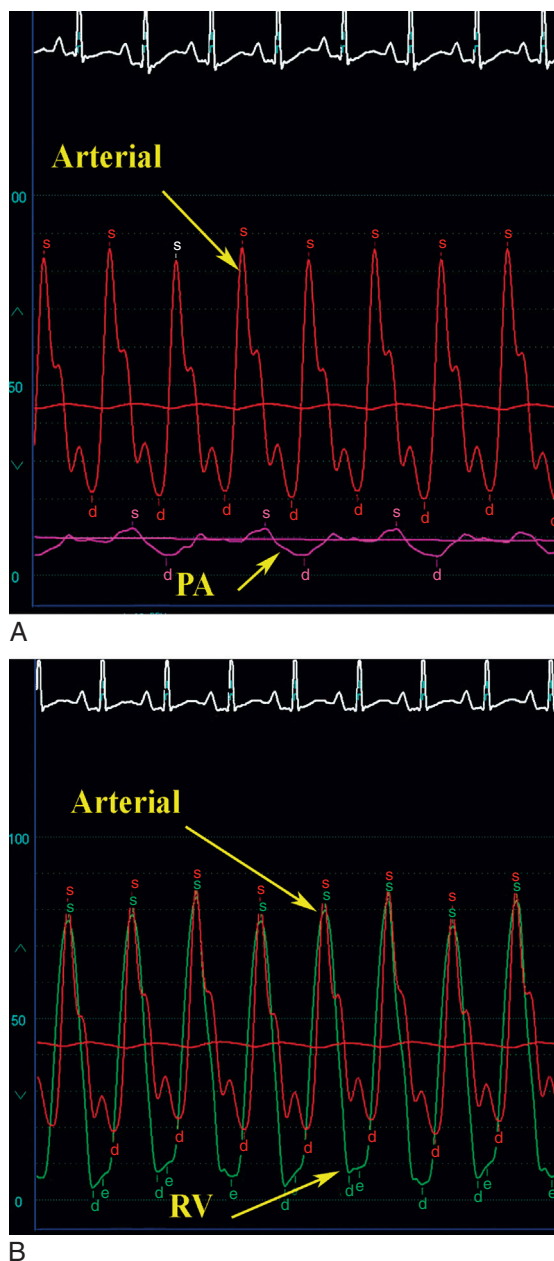
if the patient's rhythm is atrial fibrillation. The finding of ventricular interdependence is an important clue that may distinguish these two conditions (see [Chapter 10](#)).<sup>25</sup>

Cardiac output determination using the thermodilution method may be problematic and should be avoided in patients with tricuspid regurgitation because severe degrees of regurgitation underestimate the cardiac output.<sup>26</sup> Instead, most operators consider the Fick methodology to be more accurate in this setting.

## PULMONIC VALVE STENOSIS

Pulmonic stenosis is the most common abnormality of the pulmonary valve and nearly always has a congenital cause. It may be seen in association with other congenital heart defects or exist in isolation. Often detected in childhood, pulmonic stenosis rarely presents in adults. In the majority of cases, the valve leaflets are fused and amenable to a balloon or surgical valvotomy. 10% to 15% of pulmonic valves stenosed from dysplastic conditions (as seen in association with Noonan syndrome) are often not treatable by valvotomy.

Obstruction causes a pressure gradient across the pulmonic valve, with right ventricle systolic pressure exceeding pulmonary artery systolic pressure ([Fig. 8.8](#)).<sup>27</sup> Pressure overload and subsequent hypertrophy of the right ventricle ensue. The hemodynamic abnormalities depend on the severity of stenosis and the cardiac output. In mild cases the pressure gradient across the pulmonic valve is <20 mm Hg and the cardiac output increases normally with exercise.<sup>27,28</sup> With severe pulmonic stenosis, the pressure gradient exceeds 40 mm Hg and may reach very high levels (>100 mm Hg), causing the right ventricular pressure to equal systemic arterial pressures. The right ventricular stroke volume is fixed and unable to augment with exercise.<sup>28</sup> In addition, because of diminished right ventricular compliance from concentric hypertrophy of the right ventricle, severe pulmonic stenosis elevates right ventricular end-diastolic pressure, both at rest and with exercise. An elevated right ventricular end-diastolic pressure may raise right atrial pressure and cause right-to-left shunting if there is a patent foramen ovale, leading to cyanosis or paradoxical embolism. Furthermore, right ventricular diastolic pressure rises have been associated with elevations in left ventricular end-diastolic pressures, likely from interactions via the septum.<sup>28</sup> Interestingly, many individuals are asymptomatic, even with severe stenosis. Symptoms of severe stenosis include dyspnea, fatigue, syncope, and exercise intolerance.



**Fig. 8.8** Right heart pressures obtained in an infant with severe, congenital pulmonic stenosis. (A) The systolic pulmonary artery pressure measured 15 mm Hg. (B) Right ventricular pressure reached systemic levels at nearly 80 mm Hg. *d*, Diastole; *PA*, pulmonary artery; *RV*, right ventricular; *s*, systole.

Valve area can be calculated using the Gorlin formula, as described in [Chapter 5](#), and adapted to the pulmonic valve.<sup>30</sup> However, most clinicians classify the severity of pulmonic stenosis and base treatment decisions on the extent of the transvalvular gradient alone. Current guidelines classify valvular pulmonic stenosis as mild if the peak gradient is <36 mm Hg or the Doppler velocity is less than 3 m/s, moderate if the peak gradient is between 36 and 64 mm Hg or the Doppler velocity





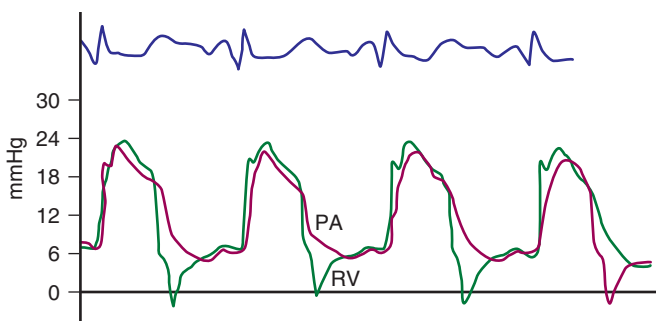


obstructs the outflow tract. With the relief of valvular obstruction, hypertrophy regresses and the extent of infundibular stenosis regresses.<sup>32</sup>

## PULMONIC VALVE REGURGITATION

Pulmonary insufficiency is uncommon. It is most often seen in association with congenital heart disease, typically because of prior surgical or balloon valvotomy for pulmonic stenosis or from the repair of tetralogy of Fallot. Other causes include rheumatic heart disease, endocarditis, dilatation of the pulmonary artery (either idiopathic or from pulmonary hypertension), traumatic disruption of the pulmonic valve, syphilis, or an isolated congenital defect.<sup>33,34</sup> The low-pressure circuit of the right heart causes pulmonary regurgitation to behave differently than aortic regurgitation.<sup>35</sup> Right atrial contraction can maintain forward pulmonary blood flow despite severe regurgitation, and the pulmonary resistance is typically very low, allowing blood to easily pass through the lungs and preventing significant backward flow during diastole. Thus the volume overload on the right ventricle is substantially less than that seen in severe aortic regurgitation, which allows this lesion to be tolerated for longer periods. Conditions that increase pulmonary vascular resistance, however, will increase the regurgitant volume and may lead to detrimental effects. Eventually, the right ventricle dilates and becomes dysfunctional, leading to reduced exercise capacity and right heart failure.

The hemodynamic abnormalities reflect the severity of pulmonic regurgitation. Patients with severe pulmonary regurgitation demonstrate an increased pulmonary arterial pulse pressure, a rapid diastolic collapse, and early equilibration of the diastolic pressures between the pulmonary artery and right ventricle or “diastasis”<sup>33,34,36</sup> (Figs. 8.10 and 8.11). The pulmonary artery pressure becomes “ventricularized” (Fig. 8.12).<sup>37</sup> Milder forms of pulmonary regurgitation affect the pulse pressure to a lesser degree, and equilibration of the pressure between the right ventricle and pulmonary artery occurs only at end diastole.



**Fig. 8.10** Diastasis of pressure between the right ventricular and pulmonary artery pressure is a hemodynamic finding of severe pulmonic insufficiency. (From Nemickas R, Roberts J, Gunnar RM, Tobin JR. Isolated congenital pulmonic insufficiency. Differentiation of mild from severe regurgitation. *Am J Cardiol.* 1964;14:456–463.) PA, Pulmonary artery; RV, right ventricular.



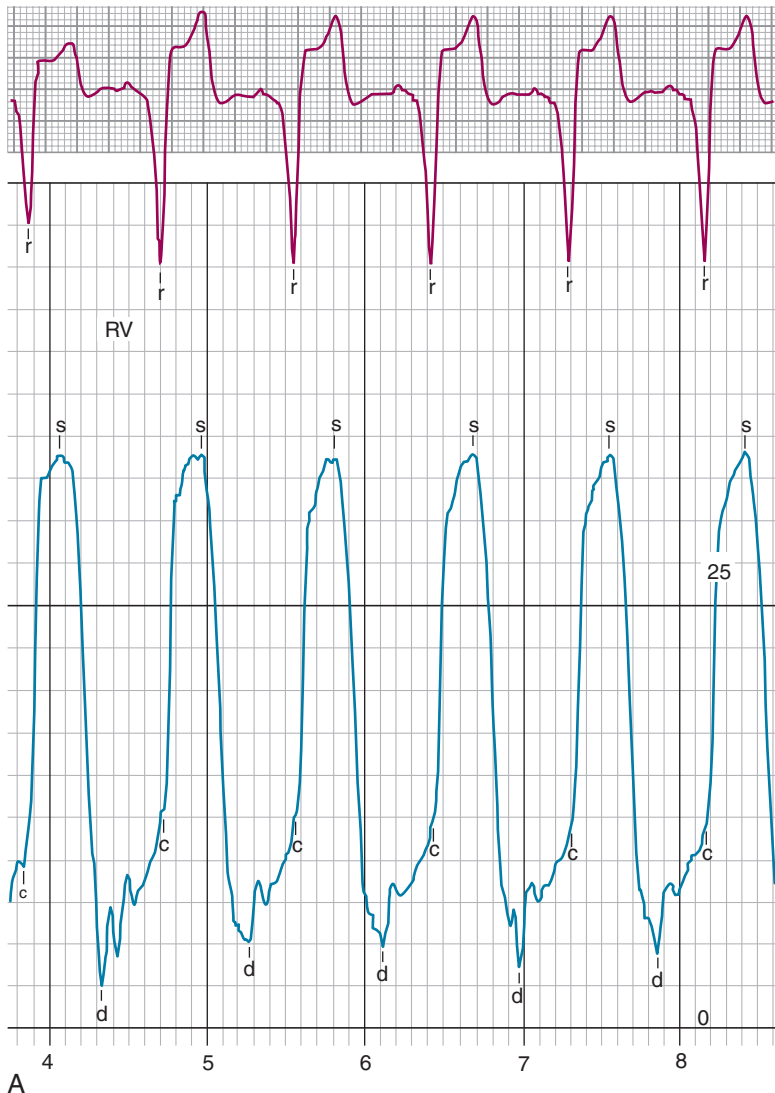
**Fig. 8.11** This is an example of severe pulmonic valve regurgitation in a 10-year-old with a history of tetralogy of Fallot with pulmonary atresia and multiple complex surgeries in the past. Shown here is diastasis between right ventricular and pulmonary artery pressures; there is also a modest gradient consistent with stenosis. *d*, Diastole; *e*, end diastole; *PA*, pulmonary artery; *RV*, right ventricular; *s*, systole.

## RIGHT VENTRICULAR FAILURE

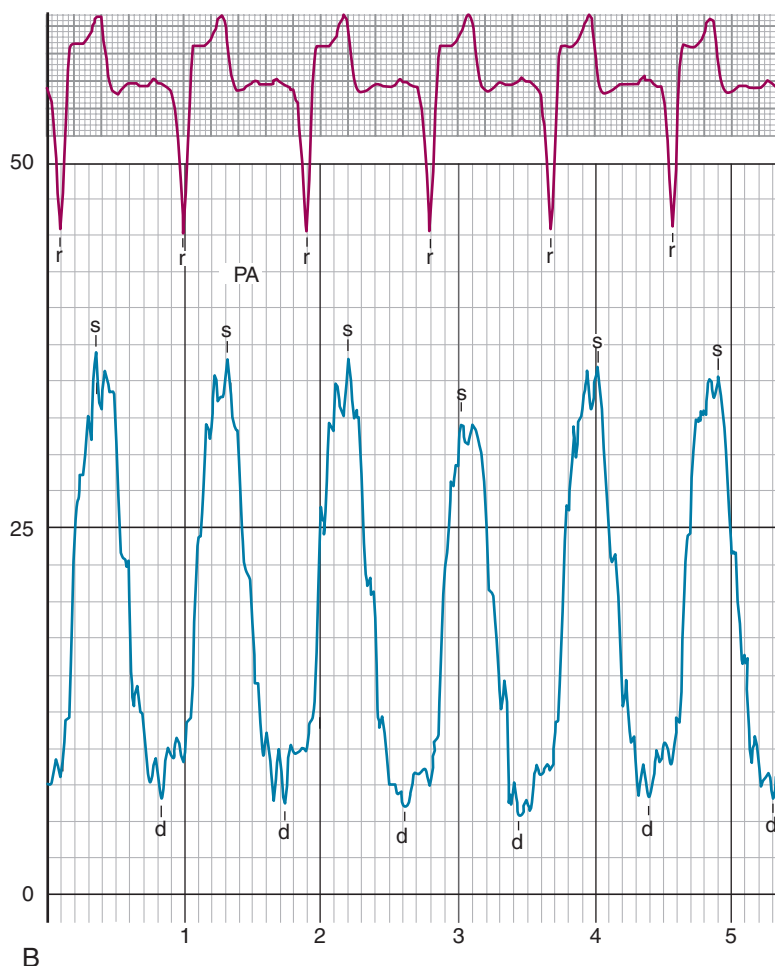
The right ventricle is subject to failure, most commonly from associated left heart failure. Isolated right heart failure can occur in a variety of settings including severe pulmonary hypertension, chronic severe pulmonic insufficiency, severe tricuspid regurgitation, right ventricular infarction, chronic, severe lung disease, acute massive pulmonary embolism, myocardial contusion from trauma, focal myocarditis, or inadequate cardiac preservation during heart surgery, or from an acute rejection after cardiac transplantation (Fig. 8.13). Several of these conditions are discussed elsewhere in this text, and the remaining discussion will focus on the hemodynamics of right ventricular infarction.

## RIGHT VENTRICULAR INFARCTION

Infarction of the right ventricle may present with dramatic hemodynamic consequences and is one of the major causes of cardiogenic shock in patients with acute myocardial infarction. Initially described as a distinct clinical entity in 1974, it usually occurs in association with between one-third and one-half of all inferior wall myocardial infarctions.<sup>38,39</sup>



**Fig. 8.12** These tracings were obtained from a patient with severe pulmonic insufficiency after surgical correction of tetralogy of Fallot. (A) The right ventricular pressure. *d*, Diastole; *c*, end-diastolic pressure; *r*, R wave; *s*, systole.

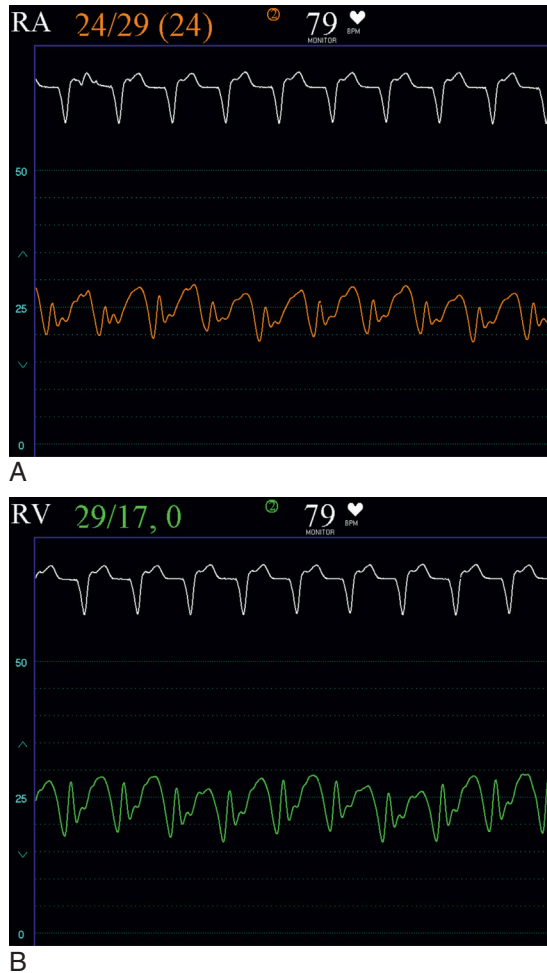


**Fig. 8.12, cont'd.** (B) The pulmonary artery pressure. Note the wide pulse pressure on the pulmonary artery tracing with equilibration of diastolic pressures. PA, Pulmonary artery; RV, right ventricular.

Occlusion of the right coronary artery proximal to the right ventricular marginal branches is the most common cause, but right ventricular infarction may also occur with occlusion of the left circumflex or even the left anterior descending in a minority of patients.<sup>39,40</sup> Rarely it may occur in isolation (i.e., without associated left ventricular infarction) from occlusion of a nondominant right coronary artery or occlusion of a right ventricular marginal branch. This may occur as an iatrogenic event from percutaneous coronary intervention of a right coronary artery if there is a loss of a right ventricular marginal branch.

Interestingly, not all cases of proximal right coronary occlusion cause right ventricular infarction. This may be explained by several mechanisms. The right ventricle is fairly resistant to ischemic injury. Oxygen demand is much lower in the right ventricle, given its smaller muscle mass. Accordingly, the susceptibility of the right ventricle to infarction and ischemia is increased in patients with right ventricular hypertrophy. Other protective mechanisms include collateral supply of the right ventricle and, possibly, perfusion of the right ventricular myocardium directly from the blood in the ventricular cavity via the Thebesian veins. Each of these factors is believed to also explain the high likelihood of recovery of right ventricular function following an infarction. Early recognition of right ventricular infarction remains important because although the long-term prognosis of right ventricular infarction is good, in-hospital morbidity and mortality are high.<sup>41–43</sup>

Even if there is a right ventricular infarction, the clinical consequences of this vary greatly, from no apparent hemodynamic abnormality to profound shock and cardiovascular collapse. This wide spectrum of clinical presentations is because the pathophysiology of right ventricular infarction depends on many complex factors (Box 8.2). Each factor plays varying roles in any given patient, explaining the variety of hemodynamic findings observed in the event of a right ventricular ischemic injury.



**Fig. 8.13** These hemodynamics were obtained in a patient with a prior heart transplant who presented with acute, severe rejection. (A) The right atrial pressure is markedly increased with a mean pressure of 24 mm Hg. (B) There is “atrialization” of the right ventricular waveform from severe right ventricular failure.

### Box 8.2 Important Variables That Impact the Pathophysiology of Right Ventricular Infarction

#### Right Ventricular Systolic Dysfunction

- Decreased left-sided preload
- Right ventricular dilatation

#### Right Ventricular Diastolic Dysfunction

- Diminished compliance of right ventricle
- Elevated filling pressures

#### Pericardial Constraint

- Increases right-sided filling pressures
- Enhances systolic ventricular interactions via the septum

#### Systolic Ventricular Interactions via the Septum

- Impaired left ventricular function
- Dependence on left ventricular function to enhance pulmonary blood flow

#### Right Atrial Function

- Maintains preload
- Loss of atrial function is more important than other infarctions

The pathophysiology of a right ventricular infarction is complex. Acute ischemia of the right ventricle results in profound *systolic dysfunction*, which decreases right ventricular stroke volume and peak systolic pressure and thus reduces left ventricular preload. This causes a drop in cardiac output and hypotension. The right ventricle acutely dilates. Acute ischemia also results in *diastolic dysfunction*. Ischemia causes the right ventricle to become stiff, and this elevates right-sided filling pressures during diastole and increases resistance to early filling. The pericardial space becomes filled with the acutely dilated right ventricle, which increases intrapericardial pressure, further impairing right ventricular and left ventricular filling. Furthermore, the effect of *pericardial constraint* also facilitates *systolic ventricular interactions mediated by the septum* by shifting the interventricular septum toward the preload-deprived left ventricle, further decreasing left ventricular filling and cardiac output. However, left ventricular contraction may cause septal bulging to the right, which may be sufficient to generate or augment right ventricular systolic force and enhance pulmonary blood flow. Thus if an associated large left ventricular infarction is present, this force may be lost and cause further worsening of hemodynamics. Finally, the *right atrium* plays an extremely important role in maintaining adequate right ventricular preload. If there is an associated right atrial infarction present, or, if there is loss of atrioventricular (AV) synchrony from the development of heart block or atrial fibrillation, significant hemodynamic deterioration may ensue.

The location of the coronary occlusion leading to right ventricular infarction is an important determinant of the clinical consequences observed. If occlusion is proximal to the right ventricular marginal branches but distal to the right atrial branches, then the main effect is a decrease in right ventricular systolic function. Right atrial contraction is augmented, and the preserved right atrial contraction can enhance right ventricular performance. The right atrial mean pressure is elevated but with an amplified “a” wave creating a characteristic “W” pattern on the right atrial waveform.<sup>44</sup> If, however, occlusion is proximal to the right atrial branches, then loss of right atrial function also occurs, and without the assistance from atrial contraction the impaired right ventricular systolic function is poorly tolerated. The right atrial pressure becomes markedly elevated with a diminished or absent “a” wave creating a characteristic “M” pattern on the right atrial waveform.<sup>44</sup>

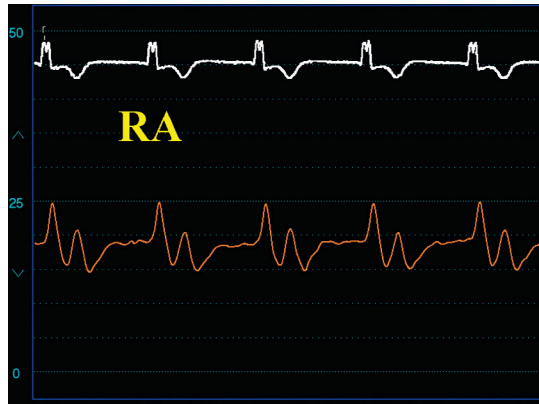
The net effect of right ventricular infarction is that left-sided filling pressure may be low, despite elevated right-sided pressure, which is clinically apparent by the triad of hypotension, clear lung fields, and elevated jugular venous pressure. The physical exam may also reveal the Kussmaul sign (paradoxical distension of the neck veins with inspiration), a feature that is both sensitive and specific for right ventricular infarction.<sup>45</sup> The physical findings may be confused with pericardial disease because right ventricular infarction mimics both tamponade and constrictive pericarditis.

Possible hemodynamic findings of right ventricular infarction include elevated right atrial pressure, typically exceeding the pulmonary capillary wedge pressure. The right atrial pressure often exceeds 10 mm Hg, and in cases with marked hemodynamic compromise, the ratio of right atrial pressure to pulmonary capillary wedge pressure is  $>0.8$ .<sup>38,46</sup> These hemodynamic findings may be masked by intravascular volume depletion and only emerge with volume loading.<sup>47</sup> The condition in which right atrial pressure exceeds left atrial pressure may cause right-to-left shunting in the presence of a patent foramen ovale.<sup>44</sup> This scenario should be suspected in the event of unexplained hypoxia not corrected with oxygen administration in the setting of right ventricular infarction; however, it is very rare and has probably appeared more often on board examination questions than in patients.

Elevated right atrial pressure is primarily due to right ventricular diastolic dysfunction, but pericardial constraint and right ventricular failure also contribute. Impaired filling of the right ventricle is evidenced by a blunted y descent on the right atrial waveform (Fig. 8.14). The a wave on the right atrial tracing reflects the strength of contraction in the right atrium. In right ventricular infarction, right atrial contraction is enhanced by the increased preload, resulting in augmentation of the height of the a wave. The x descent may be steep because of enhanced atrial relaxation. This may be seen as a W pattern on the right atrial waveform (Fig. 8.15).<sup>44</sup> These features benefit right ventricular filling. However, if an infarction involves the right atrium, then the a wave and x descents are depressed. This produces a



**Fig. 8.14** Right atrial waveform from a patient with right ventricular infarction, demonstrating a large a wave with prominent x descent and a blunted y descent. a, a wave; RA, right atrial; y, y descent.

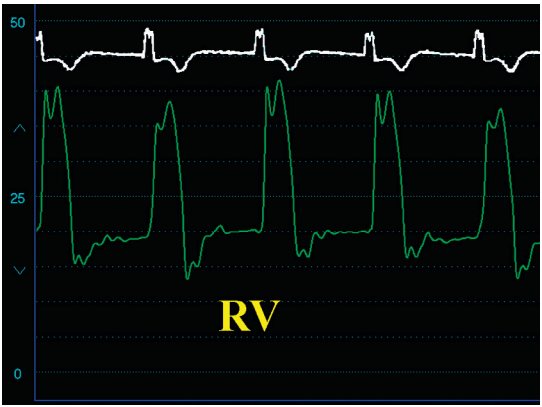


**Fig. 8.15** Example of the *W* pattern seen on a right atrial waveform in a patient with right ventricular infarction. Note the prominent *a* wave. RA, Right atrial.

characteristic *M* pattern.<sup>44</sup> Thus two hemodynamic subtypes exist based on the status of the right atrium, with both having a relatively blunted *y* descent. The first, or *W* pattern, is usually associated with right coronary artery occlusion proximal to the right ventricular branches but distal to the right atrial branches and is associated with better hemodynamics than the second, or *M* pattern, which is associated with occlusion of the right coronary artery proximal to the right atrial branches and worse hemodynamics. Although the hemodynamics are better in those with a *W* pattern than those with an *M* pattern, those with a *W* pattern can decompensate quickly when they lose AV synchrony.

Right ventricular systolic dysfunction results in a right ventricular pressure tracing characterized by a broad upstroke that reflects a depressed upstroke, reduced peak pressure, and delayed relaxation. The diastolic portion of the right ventricular pressure curve has been described as a “dip and plateau” that reflects decreased compliance as well as the constraining forces of the pericardium (Fig. 8.16). This constraint imparted by the pericardium may also produce equalization of left ventricular and right ventricular diastolic pressures, similar to constrictive pericarditis (Fig. 8.17).

The treatment of right ventricular infarction includes prompt reperfusion, maintenance of right ventricular preload, reduction of right ventricular afterload, and inotropic support of the failing ventricle. The benefit of reperfusion in the setting of right ventricular infarction has not been well defined, but several studies have shown lower rates of right ventricular infarction and more rapid recovery in right ventricular function in patients who undergo successful reperfusion of the infarct-related artery.<sup>41,48</sup> In addition to prompt reperfusion of the infarct-related artery, it is also worthwhile to try to open the occluded right ventricular marginal branches in the setting of acute right ventricular infarction and shock, as this may lead to rapid improvements in hemodynamics. Because of preload dependence, volume infusion raises the blood pressure and cardiac output associated with right ventricular infarction and often requires several liters of normal saline. Once a patient is adequately volume loaded (right atrial pressure of 15–20 mm Hg), however, there is a little additional benefit from volume infusion in improving cardiac output; more aggressive infusion of volume may lead to pulmonary edema, particularly if there is a large, associated left ventricular infarction that impairs left ventricular function. It is prudent to guide volume infusion in a critically ill patient with right ventricular infarction with an indwelling right heart catheter.<sup>49</sup> Recognition of preload dependence is important because many of the drugs typically used to treat patients with acute myocardial infarction, such as nitrates or diuretics, may produce deleterious hemodynamic effects if right ventricular infarction is also present. If volume replacement fails to improve cardiac output and blood pressure, then inotropic support is necessary. Norepinephrine is often the first-line drug for the treatment of hypotension. Dobutamine and milrinone are also helpful in improving forward flow by reducing pulmonary vascular resistance and therefore right ventricular afterload. Right ventricular afterload may also be increased when left ventricular dysfunction accompanies right ventricular infarction and elevates pulmonary venous pressures. In this setting, afterload-reducing drugs such as nitroprusside or insertion of an intraaortic balloon pump may be beneficial. Maintenance of AV synchrony is extremely important because, as mentioned earlier, the loss of atrial contraction may have deleterious hemodynamic consequences in right ventricular infarction. Cardioversion of atrial fibrillation should be performed promptly at the first sign of hemodynamic deterioration and AV pacing considered if heart block develops. If reperfusion therapy, volume replacement, and parenteral inotropic support are inadequate, mechanical circulatory support of the right ventricle may be indicated. The use of an indwelling pulmonary artery pressure catheter can help guide this decision.<sup>49</sup> As mentioned earlier, the ratio of mean right atrial pressure to mean pulmonary capillary wedge pressure greater than 0.8 is associated with right ventricular failure. In addition, the calculation of the pulmonary artery pulsatility index or PAPI can help determine which patients need mechanical circulatory support. The PAPI is calculated as the ratio of the pulmonary artery pulse pressure (calculated as pulmonary artery systolic pressure minus pulmonary artery diastolic pressure) divided by the mean right atrial pressure. A PAPI <1.0 is associated with right ventricular failure.<sup>49</sup> There are several percutaneous methods to support the failing right ventricle available, and all are able to effectively support the right heart and improve hemodynamics.<sup>49–51</sup> An example of hemodynamics obtained in a patient with profound



**Fig. 8.16** Right ventricular waveform in a patient with right ventricular infarction, demonstrating the *square root sign* or *dip and plateau* during diastole. *RV*, Right ventricular.



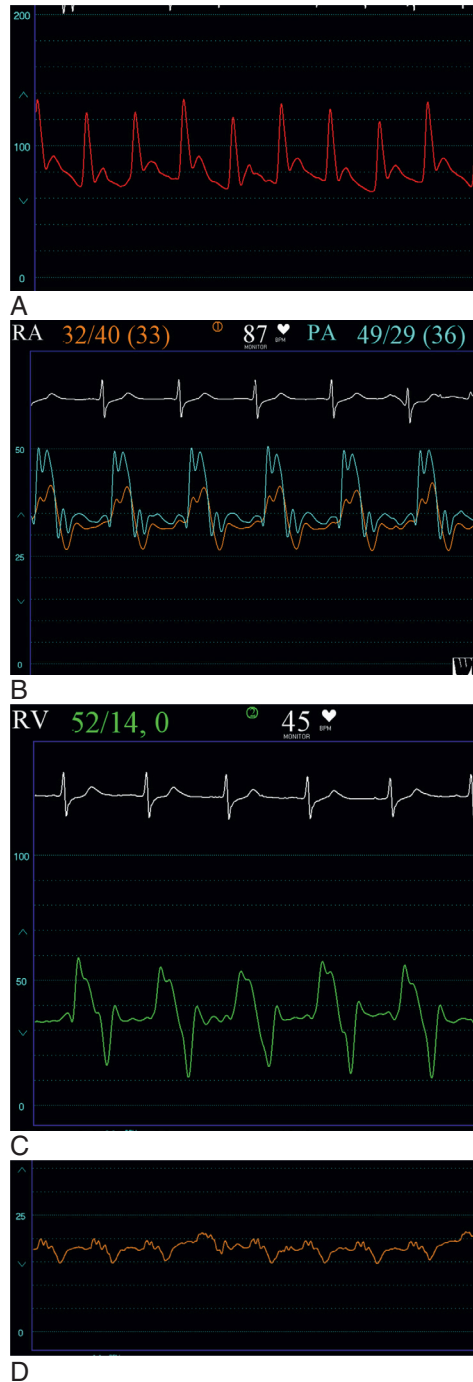
A



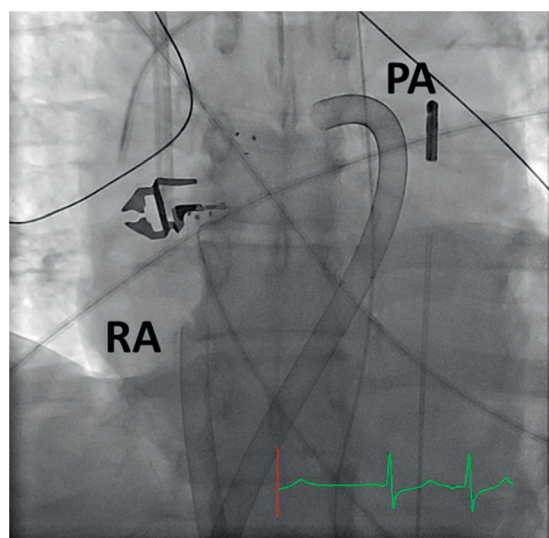
B

**Fig. 8.17** (A and B) Examples of equalization of the right ventricular and left ventricular diastolic pressures or *pseudoconstriction* pattern. (B) Note that the early left ventricular diastolic pressure is lower than the right ventricular diastolic pressure due to impaired left ventricular filling and reduced preload. Green tracing is the right ventricular waveform, and yellow tracing is the left ventricular waveform.

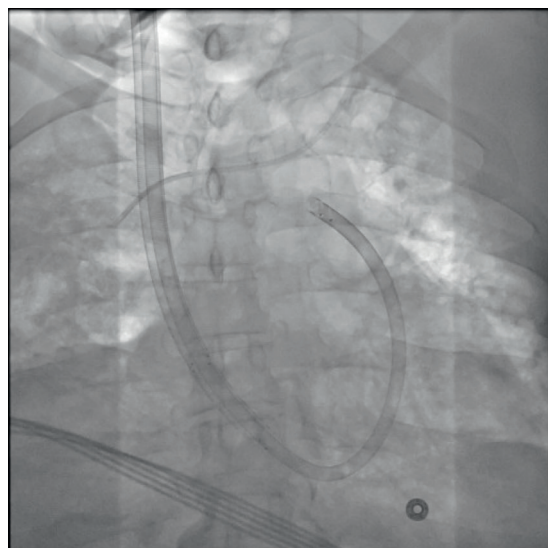




**Fig. 8.18** These hemodynamics were obtained in a patient with cardiogenic shock caused by a right ventricular infarction who was subsequently treated with a percutaneous right ventricular support device. (A) The arterial pressure waveform is consistent with shock and shows a very narrow pulse width. (B) There is marked elevation of the right atrial pressure (mean 33 mm Hg) and right ventricular diastolic pressure. (C) Right ventricular pressure waveform shows marked elevation of the right ventricular diastolic pressure with a dip and plateau pattern. (D) Right atrial pressure waveform after insertion of the percutaneous right ventricular assist device with reduction of pressure to a mean of 15–18 mm Hg.



**Fig. 8.19** TandemHeart (LivaNova) configured to support the right heart. A 21-French cannula is inserted via the femoral vein into the pulmonary artery, and another catheter is inserted via the opposite femoral vein into the right atrium. The device pulls from the right atrium and pumps into the pulmonary artery, allowing support for the failing right ventricle. PA, Pulmonary artery; RA, right atrium.



**Fig. 8.20** Protek Duo catheter (LivaNova), a dual-lumen catheter available in 29- or 31-French sizes designed for support of the right heart. One port of the catheter pulls blood from the right atrium while another port pumps blood to the pulmonary artery.

shock from a right ventricular infarction is shown in Fig. 8.18. Note the marked elevation in right atrial pressure and the calculated PAPI was 0.6, indicating marked right ventricular failure. In this case, the pulmonary artery saturation increased from 40% to 59% and the mean right atrial pressure decreased from 33 to 15 mm Hg after the insertion of a TandemHeart device (CardiacAssist, Pittsburgh, Pennsylvania) configured to support the right side (Fig. 8.19) with one catheter placed in the right atrium and another in the pulmonary artery. Dedicated catheters have been developed to allow percutaneous mechanical support of the right side of the heart.<sup>49</sup> The ProtekDuo (LivaNova, Arvada, Colorado) is a dual-lumen catheter developed for use with the TandemHeart system (Fig. 8.20). There is also the IMPELLARP device (Abiomed, Inc., Danvers, Massachusetts); extracorporeal membrane oxygenation may also be used if there is associated hypoxemia.

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